SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS

Course Code:	**CSA1205**	**Course Title:**	**Computer Architecture for Multicore Systems**
CO Assessed:	**CO4**	**Assessment:**	**ASSESSMENT TOOL3- COMPARATIVE ANALYSIS**
Weightage:	**30%**	**Total Marks:**	**25**
CO4 Statement:	*Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.*		
Student Name:	**SHAIK HAZIPEERA**	**Reg.No.:**	**192572056**
Department:	**CSE(AI)**	**Date:**	**07-08-2026**

ANNEXURE A
COMPARATIVE ANALYSIS

1.Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.
2.Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.
3.Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.
4.Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.
5.Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Code of Conduct:

I SHAIK HAZIPEERA (192572056) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

S. Hazipeera
Signature of the student:

MARKS ALLOCATION

	Scope of Items (20%)	Comparison Depth (25%)	Use of Evidence (20%)	Critical Thinking (20%)	Clarity of Presentation (15%)
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

ANSWER

1. Compare L1, L2, and L3 Cache Levels in Terms of Latency, Bandwidth, and Throughput, and Analyze Their Impact on Processor Performance

Introduction

Cache memory is a high-speed memory placed between the CPU and the main memory (DRAM). It stores frequently used data and instructions to reduce memory access time. Modern processors use a **three-level cache hierarchy (L1, L2, and L3)** to improve performance.

Comparison of L1, L2, and L3 Cache

Parameter	L1 Cache	L2 Cache	L3 Cache
Location	Inside each CPU core	Inside each core	Shared among all CPU cores
Size	32–64 KB	256 KB–1 MB	2–64 MB
Latency	Lowest (1–2 cycles)	Medium (5–15 cycles)	Highest among caches (20–50 cycles)
Bandwidth	Highest	High	Moderate
Throughput	Very High	High	Moderate
Speed	Fastest	Faster	Fast

Impact on Processor Performance

L1 Cache

- Provides the fastest access to frequently used instructions and data.
- Minimizes CPU waiting time.
- Offers the **highest bandwidth and throughput**.

L2 Cache

- Acts as a backup for L1 cache.
- Reduces cache misses when data is unavailable in L1.
- Balances speed and storage capacity.

L3 Cache

- Shared by all processor cores.
- Stores commonly used data for multiple applications.
- Improves multitasking and reduces access to slower DRAM.

Advantages

- Reduces memory access latency.
- Improves processor throughput.
- Decreases cache misses.
- Enhances multitasking performance.
- Reduces frequent access to DRAM, saving power.

2. Differentiate SRAM and DRAM by Evaluating Their Latency, Bandwidth, and Throughput Characteristics in Cache and Main Memory Design

Introduction

SRAM (Static Random Access Memory) and DRAM (Dynamic Random Access Memory) are two important memory technologies used in computer systems. **SRAM** is mainly used for **cache memory**, while **DRAM** is used as **main memory (RAM)**. They differ in terms of latency, bandwidth, throughput, cost, and power consumption.

Comparison of SRAM and DRAM

Parameter	**SRAM**	**DRAM**
Full Form	Static Random Access Memory	Dynamic Random Access Memory
Speed	Very Fast	Slower than SRAM
Latency	Very Low (1–5 ns)	Higher (50–100 ns)
Bandwidth	High	Moderate to High
Throughput	Very High	High
Storage Capacity	Small	Large
Cost	Expensive	Less Expensive
Power Consumption	Higher	Lower
Used In	L1, L2, L3 Cache	Main Memory (RAM)

Role in Cache and Main Memory Design
SRAM

- Used in **L1, L2, and L3 cache** because of its very low latency.
- Provides high bandwidth and throughput for quick CPU access.
- Improves processor performance by reducing memory access time.

DRAM

- Used as **main memory** due to its large storage capacity.
- Stores the operating system, applications, and user data.
- Provides more memory at a lower cost but has higher latency than SRAM.

Advantages
SRAM

- Very low latency.
- Highest throughput.
- Faster processor performance.
- No refresh operation required.

DRAM

- Large storage capacity.
- Lower cost per bit.
- Suitable for storing large programs and data.
- Lower power consumption for large memory sizes.

3. Compare Virtual Memory with Paging and Cache Memory Mechanisms in Terms of Access Latency and System Throughput

Introduction
Virtual memory with paging and cache memory are important memory management techniques used to improve system performance. **Cache memory** provides fast access to frequently used data, while **virtual memory with paging** extends the available memory by using secondary storage (SSD/HDD) when RAM is full.

Comparison of Virtual Memory (Paging) and Cache Memory

Parameter	Cache Memory	Virtual Memory (Paging)
Purpose	Speeds up CPU access to data	Extends main memory using disk storage
Memory Used	SRAM (L1, L2, L3 Cache)	DRAM + SSD/HDD
Access Latency	Very Low (1–50 CPU cycles)	Very High due to page faults
Bandwidth	High	Low
Throughput	Very High	Lower than cache memory
Performance	Improves processor speed	Prevents memory shortage but may reduce performance
Cost	Expensive	Less Expensive

Impact on System Performance

Cache Memory

- Stores frequently accessed instructions and data.
- Reduces CPU waiting time.
- Provides very low latency and high throughput.
- Improves overall processor performance.

Virtual Memory with Paging

- Allows execution of large programs even when RAM is limited.
- Moves inactive pages between RAM and SSD/HDD.
- Page faults increase memory access time and reduce throughput.
- Useful for increasing memory capacity but slower than cache memory.

Advantages

Cache Memory

- Very low access latency.
- High bandwidth and throughput.
- Faster program execution.
- Reduces CPU idle time.

Virtual Memory (Paging)

- Supports larger applications than physical RAM.
- Improves memory utilization.
- Allows multiple programs to run simultaneously.
- Reduces the need for additional RAM.

4. **Analyze the Differences Between NAND Flash and DRAM with Respect to Latency, Bandwidth, and Suitability for Hierarchical Memory Systems**

Introduction

NAND Flash and **DRAM** are widely used memory technologies in modern computer systems. **NAND Flash** is a non-volatile memory used for permanent storage, whereas **DRAM** is a volatile memory used as the main memory (RAM). They differ in terms of latency, bandwidth, throughput, and their role in the memory hierarchy.

Comparison of NAND Flash and DRAM

Parameter	**NAND Flash**	**DRAM**
Type	Non-Volatile Memory	Volatile Memory
Speed	Slower	Faster
Access Latency	High	Low
Bandwidth	Moderate	High
Throughput	Moderate	High
Storage Capacity	Very Large	Medium to Large
Power Consumption	Lower (retains data without power)	Higher (requires continuous power)
Used In	SSDs, USB drives, Memory Cards	Main Memory (RAM)

Role in Hierarchical Memory Systems

NAND Flash

- Used for permanent storage of the operating system, applications, and user data.
- Retains data even when power is turned off.
- Suitable for secondary storage due to its high capacity and low cost.

DRAM

- Used as the main memory for running programs and the operating system.
- Provides fast access to data required by the CPU.
- Acts as an intermediate layer between cache memory and storage devices.

Advantages

NAND Flash

- Non-volatile (data is retained without power).
- Large storage capacity.
- Low power consumption.
- Cost-effective for mass storage.

DRAM

- Low access latency.
- High bandwidth and throughput.
- Faster program execution.
- Ideal for multitasking and real-time applications.

5. Compare MRAM and RRAM Technologies by Examining Their Performance in Terms of Access Time, Throughput, and Energy Efficiency

Introduction

MRAM (Magnetoresistive Random Access Memory) and **RRAM (Resistive Random Access Memory)** are emerging **non-volatile memory technologies**. Both retain data even when power is turned off and are considered promising alternatives to traditional memory due to their fast speed, low power consumption, and high reliability.

Comparison of MRAM and RRAM

Parameter	**MRAM**	**RRAM**
Full Form	Magnetoresistive RAM	Resistive RAM
Access Time	Very Fast	Fast
Throughput	High	High
Energy Efficiency	Low power consumption	Very low power consumption
Data Retention	Non-volatile	Non-volatile
Endurance	Very High	High
Storage Density	Moderate	High
Applications	Cache memory, Embedded systems	AI devices, IoT, Edge computing

Performance Analysis

MRAM

- Provides **very fast access time** and **high throughput**.
- Consumes less power than SRAM and DRAM.
- Has excellent endurance, making it suitable for frequent read/write operations.
- Commonly used in embedded systems and cache memory.

RRAM

- Offers **fast access** with **very low energy consumption**.
- Provides high storage density, allowing more data to be stored in less space.
- Suitable for AI accelerators, IoT devices, and edge computing.
- Reduces power consumption while maintaining good performance.

Advantages

MRAM

- Very low latency.
- High throughput.
- Excellent durability and reliability.
- Non-volatile with fast read/write operations.

RRAM

- Extremely energy efficient.
- High storage density.
- Fast data access.
- Suitable for low-power and AI-based applications.